

UNIFORM LOCAL TABULARITY IN INTUITIONISTIC LOGIC

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Semantic complexity

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3. **Algebras:** Very complicated!

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In the lower levels, there are several open problems, including the so-called *Kuznetsov problem* on topological completeness: whether topological completeness is equivalent to algebraic/general completeness.

On the upper levels, the boundary of tabular logics is well-understood, and has an elegant solution.

Theorem (Esakia and Grigolia, 1977)

There exist three superintuitionistic logics C_1, C_2, C_3 which are finitely axiomatizable and locally tabular (hence decidable) such that $L \in \text{Ext}(\text{IPC})$ is not tabular if and only if $L \subseteq C_i$ for some i .

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Such results exist likewise for the modal logic S4.

Question (Due to several people)

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Upshot: local tabularity is decidable above $S4$.

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4. (Hyttinen, Quadrellaro, Martins, Moraschini, 2023): there exist counterexamples.

Definition

Given a formula ϕ , the **implication-depth** $d(\phi)$:

1. $d(p) = 0 = d(\perp) = d(\top)$;
2. $d(\phi \wedge \psi) = d(\phi \vee \psi) = \max(d(\phi), d(\psi))$;
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Definition

A logic L will be said to be **n -uniform** if each formula $\phi(\bar{p})$ is equivalent to a formula over $\bar{p}$ with implication depth at most n . It will be called **uniformly locally tabular** if it is n -uniform for some n .

Formula depth and bisimulation games

Recall **bounded bisimulations**:

Definition

Let $(P, \leq, c, x)$ and $(Q, \leq, c', y)$ be two models over n . We say that a relation $S_k \subseteq P \times Q$ is a *k-bisimulation* based on (x, y) if there are relations $S_k \subseteq \cdots \subseteq S_j \subseteq \cdots \subseteq S_0 \subseteq P \times Q$ for each $0 \leq j \leq k$, such that $xS_k y$ and for each $w \in P, w' \in Q$:

1. Whenever $wS_0 w'$ then $c(w) = c'(w')$;
2. For each $j < k$, whenever $wS_{j+1} w'$ and $w \leq v$ there is some $w' \leq v'$ such that $vS_j v'$.
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$\mathfrak{M}, x \leq_k \mathfrak{N}, y$ if and only if: $k = 0$ then $x \Vdash p$ implies $y \Vdash p$; for $k > 0$, there exists $x' \geq x$ such that $\mathfrak{M}, x' \sim_{k-1} \mathfrak{N}, y$.

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The following was observed by Shehtman, who studied the concept of uniform local tabularity:

Proposition

Let $\mathbf{K}$ be a class of finite posets. Then $L = \text{Log}(\mathbf{K})$ is $n + 1$ -uniformly locally tabular if and only if for each $\bar{p}$, whenever (P, x) and (Q, y) are two models in $\mathcal{M}_{\bar{p}}(\mathbf{K})$ and $(P, x) \leq_{n+1} (Q, y)$, then $(P, x) \leq_{\infty} (Q, y)$.

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Note: the latter condition holds if n -bisimilarity implies full bisimilarity.

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We will confirm Shehtman's conjecture, exhibiting a logic which is **locally tabular** but not **uniformly so**.

Hierarchy of locally tabular logics

Some initial results which give the basic idea of the techniques at play:

Theorem

For L a superintuitionistic logic, we have:

1. *L is 0-uniform if and only if L is inconsistent.*
2. *L is 1-uniform if and only if $L = \text{CPC}$;*
3. *L is 2-uniform if and only if $L \supseteq \text{LC}$.*

Proof.

CPC is easily seen to be 1-uniform. It is covered by Sme – the logic of the two element poset. □

Stratifying locally tabular logics

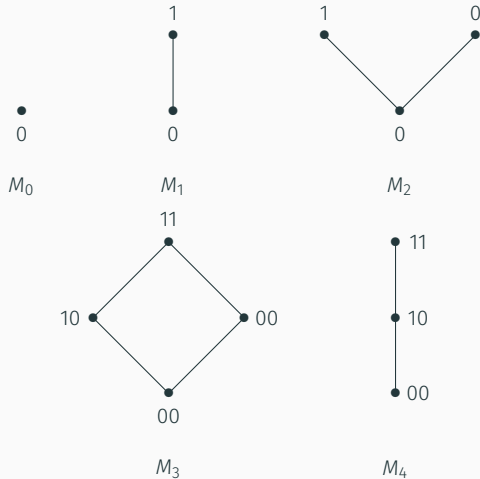


Figure 1: Models witnessing degrees of uniformity

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A finite (rooted) poset $(P, \leq)$ is called a **Boolean sum** if P has depth n , and:

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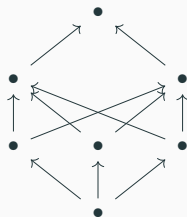


Figure 2:

We consider the following logic of **Weak Peirce's law**:

$$\text{wPL} := (q \rightarrow p) \vee (((p \rightarrow q) \rightarrow p) \rightarrow p).$$

It was shown by Esakia (1977) and independently by Mardaev (1991) that wPL is locally tabular.

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Proposition (Esakia, 1977)

A finite rooted frame P is a Boolean sum if and only if $P \Vdash \text{wPL}$.

Definition

Let P be a Boolean sum of depth n . For each n , let c_n be the cardinality of the set of elements of depth n . Given $\langle c_1, \dots, c_n \rangle$ we say that a consecutive subsequence $c_i, \dots, c_{i+k}$ where $c_{i+j} > 1$ for $0 \leq j \leq k$ is a k -stack. We say that P has *stack-depth* k if there exists a k -stack but not a $k+1$ -stack in P .

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A Boolean sum P has stack-depth 1 if and only if it does not contain Q_4 or Q_5 as a p -morphic image of a rooted upset.

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Theorem

The logic 3Uni is the smallest 3-uniform logic.

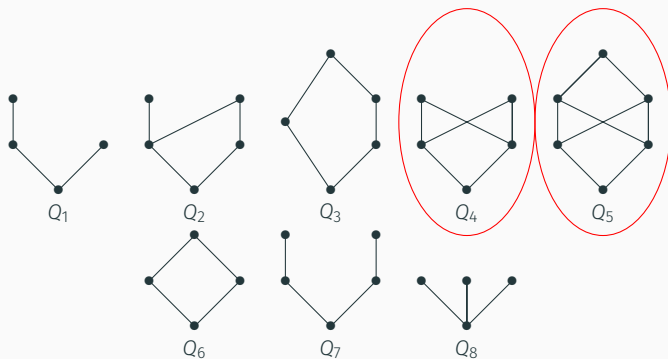


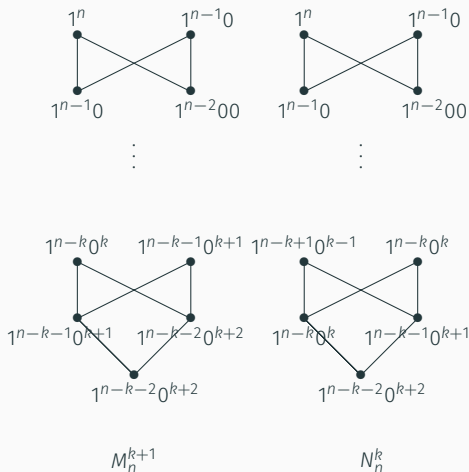
Figure 3: Q_i posets

A locally tabular but not uniformly locally tabular logic

The notion of stack-depth turns out to control, in extensions of wPL of width 2, the degree of uniformity.

Theorem

The logic wPL is locally tabular but not uniformly locally tabular.



Further remarks

The above separation allows us to reframe the problems about local tabularity on the level of uniform local tabularity.

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The logics used in separating local tabularity and uniform local tabularity appear to have some **geometric content**: the poset P which consists of exactly one k -stack is a **standard representation of the CW-complex of a sphere**.

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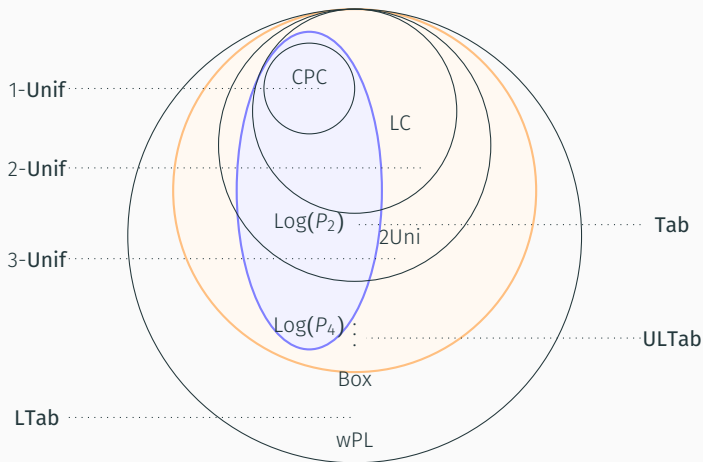


Figure 5: Hierarchy of Locally Tabular Superintuitionistic Logics; blue oval covers the tabular logics.

Thank you!
Questions?